

Technical Document #165: Software Engineering - Comprehensive Overview

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Continuous integration (CI) automatically builds and tests code changes. It catches integration issues early in the development process.

Design patterns provide reusable solutions to common software design problems. Singleton, factory, and observer patterns are widely used.

Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Key Takeaways:

- Test-driven development (TDD) writes tests before writing code.
- Design patterns provide reusable solutions to common software design problems.
- API design defines how software components communicate.